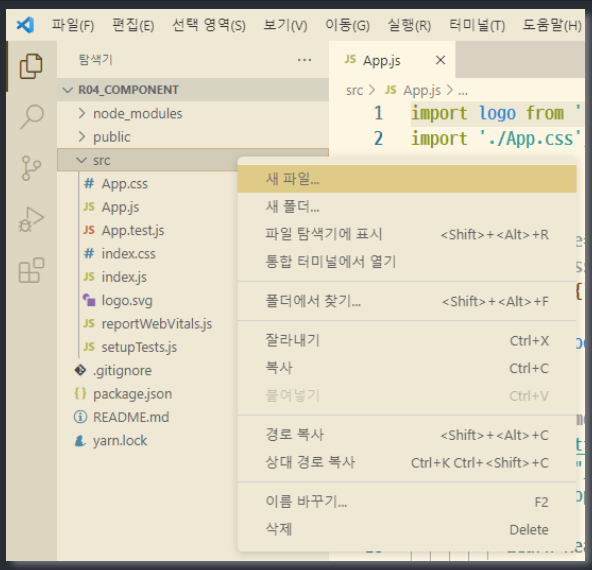
Component

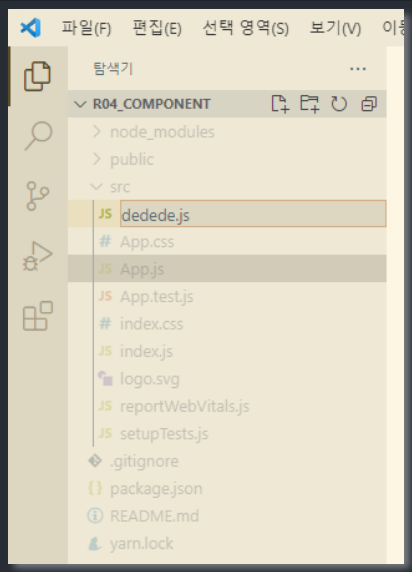
권기웅

1. 함수형 Component

1.1 Component 기본

1.1.1 파일 만들기





1.1.2 소스

dedede.js

```
import React from "react";

// rafce
const dedede = () => {
  return <div>드드드드듣듣드드드듣</div>;
};

export default dedede;
```

App.js

```
import Dedede from "./dedede";

const App = () => {
  return <Dedede />;
};

export default App;
```

1.2 props

1.2.1 props 기본

drill.js

```
import React from "react";

// racfp
const Drill = (props) => {
  return <div>드릴 {props.sound} 쯤 그만</div>;
};

export default Drill;
```

App.js

```
import Dedede from "./dedede";
import Drill from "./drill";

const App = () => {
  return (<>
    <Dedede/>
    <Drill sound="드르르르그르그륵"/>
  </>);
};

export default App;
```

1.2.2 기본값, 자료형

drill.js

```
import React from "react";
import PropTypes from "prop-types"; // imp

const Drill = (props) => {
  return (
    <div>
      {props.who} : 드릴 {props.sound} 쯤 그만
    </div>
  );
};

Drill.defaultProps = {
  sound: "드르르글륻드르륵",
};

Drill.propTypes = {
  sound: PropTypes.string, // pts
  who: PropTypes.string.isRequired, // ptsr
};

export default Drill;
```

App.js

```
import Dedede from "./dedede";
import Drill from "./drill";

const App = () => {
  return (<>
    <Dedede/>
    <Drill who="더 솔데스크"/>
  </>);
};

export default App;
```

1.2.3 텍스트

jjh.js

```
import React from "react";

const JJH = (props) => {
  return <div>장준희 : {props.children}</div>;
};

export default JJH;
```

App.js

```
import JJH from "./jjh";

const App = () => {
  return (
    <>
      <JJH>오햅니다</JJH>
    </>
  );
};

export default App;
```

1.2.4 비구조화 할당 문법

jjh.js

```
import React from "react";

const JJH = (props) => {
  const { age, children } = props; // cp
  return (
    <div>
      장준희({age}) : {children}
    </div>
  );
};

export default JJH;
```

App.js

```
import JJH from "./jjh";

const App = () => {
  return (
    <>
      <JJH age="48">오햅니다</JJH>
    </>
  );
};

export default App;
```

2. 클래스형 Component

2.1 클래스형 Component

notice.js

```
import React, { Component } from "react";

// rce
export class Notice extends Component {
  render() {
    return (
      <div>
        강의실에서 {this.props.what}할거면, 꼭 {this.props.what2}를 합시다.
      </div>
    );
  }
}

export default Notice;
```

App.js

```
import React, { Component } from "react";
import Notice from "./notice";

export class App extends Component {
  render() {
    return <Notice what="트월킹" what2="녹화" />;
  }
}

export default App;
```

3. state

3.1 클래스형 Component

3.1.1 생성자로

newFont.js

```
import React, { Component } from "react";

export class NewFont extends Component {
  constructor(props) {
    super(props);
    this.state = {
      size: 1,
    };
  }
  render() {
    const { size } = this.state; // cs
    const { comment } = this.props;
    return (
      <>
        <font size={size}>{comment}</font>
        <br></br>
        <button
          onClick={() => {
            this.setState({ size: size === 7 ? 1 : size + 1 });
          }} // anfn
        >
          커져라
        </button>
      </>
    );
  }
}

export default NewFont;
```

App.js

```
import React, { Component } from "react";
import NewFont from "./newFont";

export class App extends Component {
  render() {
    return <NewFont comment="메소같은 소리하네" />;
  }
}

export default App;
```

3.1.2 생성자 말고

```
import React, { Component } from "react";

export class NewFont extends Component {
  state = { size: 1 }; // est
  render() {
    const { size } = this.state;
    const { comment } = this.props;
    return (
      <>
        <font size={size}>{comment}</font>
        <br></br>
        <button
          onClick={() => {
            this.setState({ size: size === 7 ? 1 : size + 1 });
          }}
        >
          커져라
        </button>
      </>
    );
  }
}

export default NewFont;
```

3.1.3 비동기식이라

이래도 1씩 커짐

```
import React, { Component } from "react";

export class NewFont extends Component {
  state = { size: 1 };
  render() {
    const { size } = this.state;
    const { comment } = this.props;
    return (
      <>
        <font size={size}>{comment}</font>
        <br></br>
        <button
          onClick={() => {
            this.setState({ size: size + 1 });
            this.setState({ size: size + 1 });
            this.setState({ size: size + 1 });
          }}
        >
          커져라
        </button>
      </>
    );
  }
}

export default NewFont;
```

prevState 활용

```
import React, { Component } from "react";

export class NewFont extends Component {
  state = { size: 1 };
  render() {
    const { size } = this.state;
    const { comment } = this.props;
    return (
      <>
        <font size={size}>{comment}</font>
        <br></br>
        <button
          onClick={() => {
            this.setState((prevState, props) => {
              return { size: prevState.size + 1 };
            });
            this.setState((prevState) => {
              return { size: prevState.size + 1 };
            });
            this.setState((prevState) => ({ size: prevState.size + 1 }));
          }}
        >
          커져라
        </button>
      </>
    );
  }
}

export default NewFont;
```

3.1.4 끝나고 뭐 하기

```
import React, { Component } from "react";

export class NewFont extends Component {
  state = { size: 1 };
  render() {
    const { size } = this.state;
    const { comment } = this.props;
    return (
      <>
        <font size={size}>{comment}</font>
        <br></br>
        <button
          onClick={() => {
            this.setState({ size: size + 1 }, () => {
              alert("키웠다");
            });
          }}
        >
          커져라
        </button>
      </>
    );
  }
}

export default NewFont;
```

3.2 함수형 Component

3.2.1 Hook

newH1.js

```
import React, { useState } from "react";

const NewH1 = () => {
  const [msg, setMsg] = useState("허위정보"); // dar
  const changeMsg1 = () => setMsg("날조");
  const changeMsg2 = () => setMsg("모함");

  return (
    <>
      <h1>{msg} 대환영</h1>
      <button onClick={changeMsg1}>날조</button>
      <button onClick={changeMsg2}>모함</button>
    </>
  );
};

export default NewH1;
```

App.js

```
import React, { Component } from "react";
import NewH1 from "./newH1";

export class App extends Component {
  render() {
    return <NewH1 />;
  }
}

export default App;
```

3.2.2 여러 개

```react/NewH1.jsx
import React, { useState } from "react";

const NewH1 = () => {
  const [msg, setMsg] = useState("허위정보"); // dar
  const changeMsg1 = () => setMsg("날조");
  const changeMsg2 = () => setMsg("모함");
  const [csss, setCsss] = useState({ backgroundColor: "red", color: "white" });

  return (
    <>
      <h1>
        <span style={csss}>{msg}</span> 대환영
      </h1>
      <button
        onClick={() => {
          changeMsg1();
          setCsss({ backgroundColor: "green", color: "orange" });
        }}
      >
        날조
      </button>
      <button
        onClick={() => {
          changeMsg2();
          setCsss({ backgroundColor: "blue", color: "pink" });
        }}
      >
        모함
      </button>
    </>
  );
};

export default NewH1;
```

4. 반복문

4.1 반복문 관련 메소드

4.1.1 map

```
var ar = [1, 2, 3, 4, 5];
var ar2 = ar.map(function (val, index, arr) {
  // val : 1,2,3,4,5
  // index : 0,1,2,3,4
  // arr : [1, 2, 3, 4, 5]
  return val * index;
});
// ar2 = [0, 2, 6, 12, 20]
```

4.1.2 filter

```
var ar = [1, 2, 3, 4, 5];
var ar2 = ar.filter(v => v % 2 === 0);
// ar2 = [2, 4]
```

4.2 반복문 활용

4.2.1 가짜 게시판 준비

fakeBBS.js

```
import React, { useState } from "react";

const FakeBBS = () => {
  const [snacks, setSnacks] = useState([]);
  const snackList = snacks.map((s, i) => (
    <tr key={i}>
      <td>{s.name}</td>
      <td>{s.price}</td>
    </tr>
  ));
  return (
    <>
      <table border={1}>
        <tr>
          <th>이름</th>
          <th>가격</th>
        </tr>
        {snackList}
      </table>
    </>
  );
};

export default FakeBBS;
```

App.js

```
import React, { Component } from "react";
import FakeBBS from "./fakeBBS";

export class App extends Component {
  render() {
    return <FakeBBS />;
  }
}

export default App;
```

4.2.2 추가기능

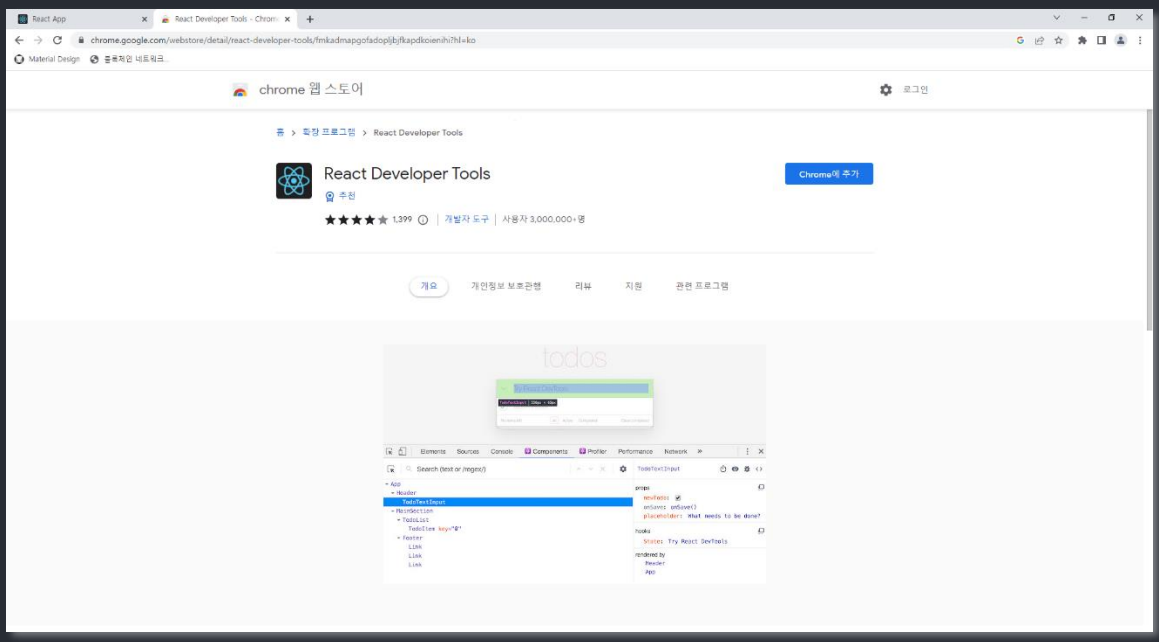
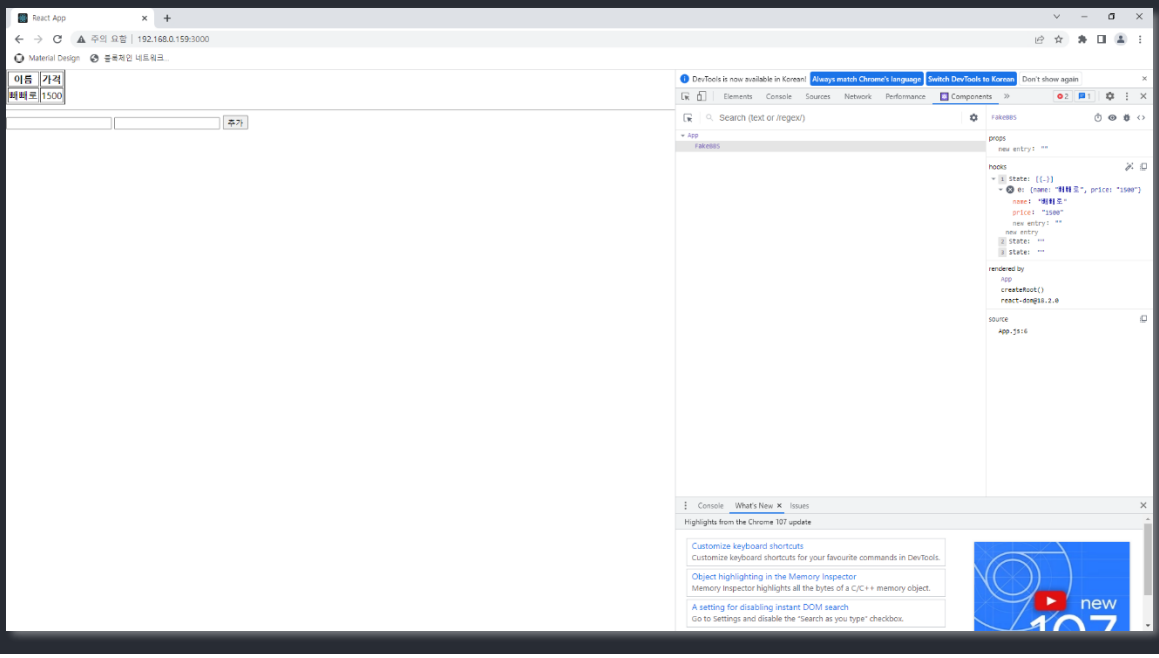
fakeBBS.js

```
import React, { useState } from "react";

const FakeBBS = () => {
  const [snacks, setSnacks] = useState([]);
  const snackList = snacks.map((s, i) => (
    <tr key={i}>
      <td>{s.name}</td>     <td>{s.price}</td>
    </tr>
  ));

  const [nameTxt, setNameTxt] = useState("");
  const [priceTxt, setPriceTxt] = useState("");
  const add = () => {
    const newSnacks = snacks.concat({ name: nameTxt, price: priceTxt });
    setSnacks(newSnacks);
    setNameTxt("");     setPriceTxt("");
  };

  return (
    <>
      <table border={1}>
        <tr>
          <th>이름</th>     <th>가격</th>
        </tr>
        {snackList}
      </table>

      <hr />
      <input value={nameTxt} onChange={(e) => { setNameTxt(e.target.value); }} /> 
      <input value={priceTxt} onChange={(e) => { setPriceTxt(e.target.value); }} /> 
      <button onClick={add}>추가</button>
    </>
  );
};

export default FakeBBS;
```

4.2.3 삭제기능

```
import React, { useState } from "react";

const FakeBBS = () => {
  const [snacks, setSnacks] = useState([]);

  const snackList = snacks.map((s, i) => (
    <tr key={i} onDoubleClick={() => { del(s.name) }}>
      <td>{s.name}</td>     <td>{s.price}</td>
    </tr>
  ));

  const [nameTxt, setNameTxt] = useState("");
  const [priceTxt, setPriceTxt] = useState("");
  const add = (e) => {…
  };

  const del = (n) => {
    const newSnacks = snacks.filter((s) =>  s.name !== n )
    setSnacks(newSnacks)
  };

  return (
    <>
      <table border={1}>
        <tr>
          <th>이름</th>     <th>가격</th>
        </tr>
        {snackList}
      </table>
      <hr />
      <input value={nameTxt} onChange={(e) => { setNameTxt(e.target.value); }} /> 
      <input value={priceTxt} onChange={(e) => { setPriceTxt(e.target.value); }} /> 
      <button onClick={add}>추가</button>
    </>
  );
};

export default FakeBBS;
```

5. chrome 확장 프로그램

5.1 React Developer Tools

React Developer Tools